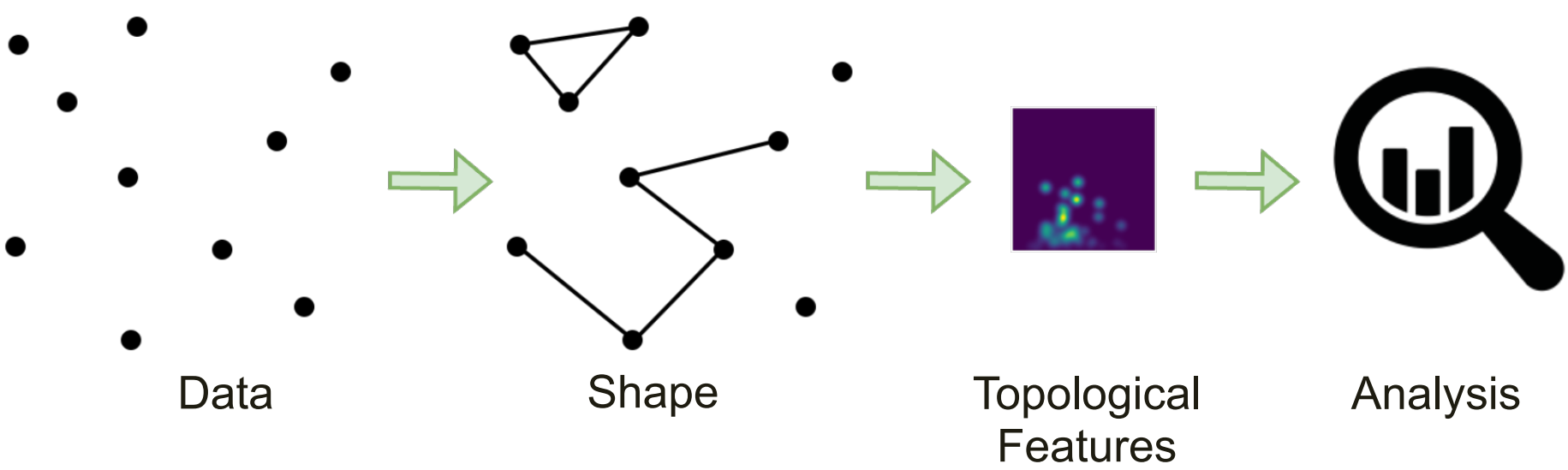


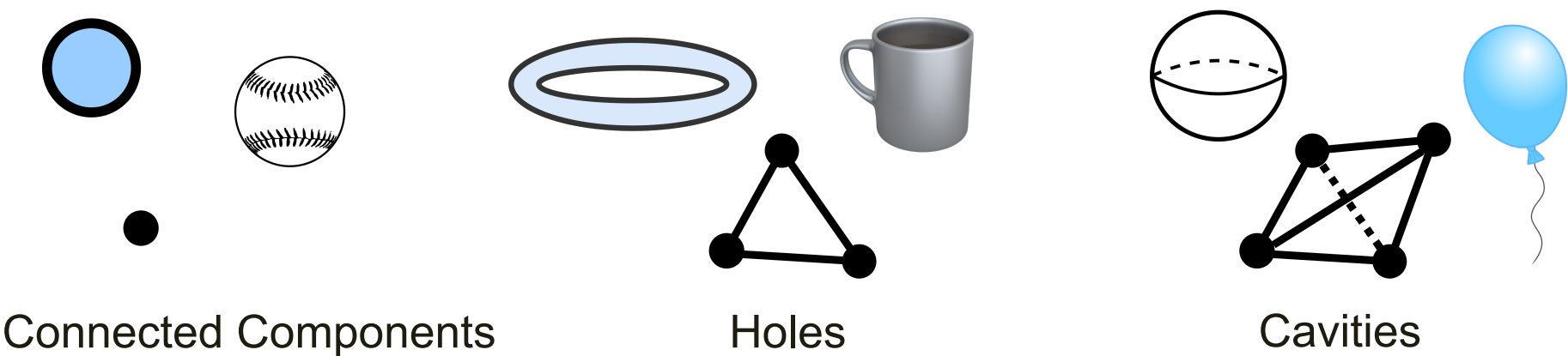
Introduction

- Goal of this project is to predict overdose risk using socioeconomic factors.
- Topological Data Analysis (TDA) can uncover hidden patterns and information from the data.
- We will use the topology of the Social Vulnerability Index (SVI) data to identify overdose risk.

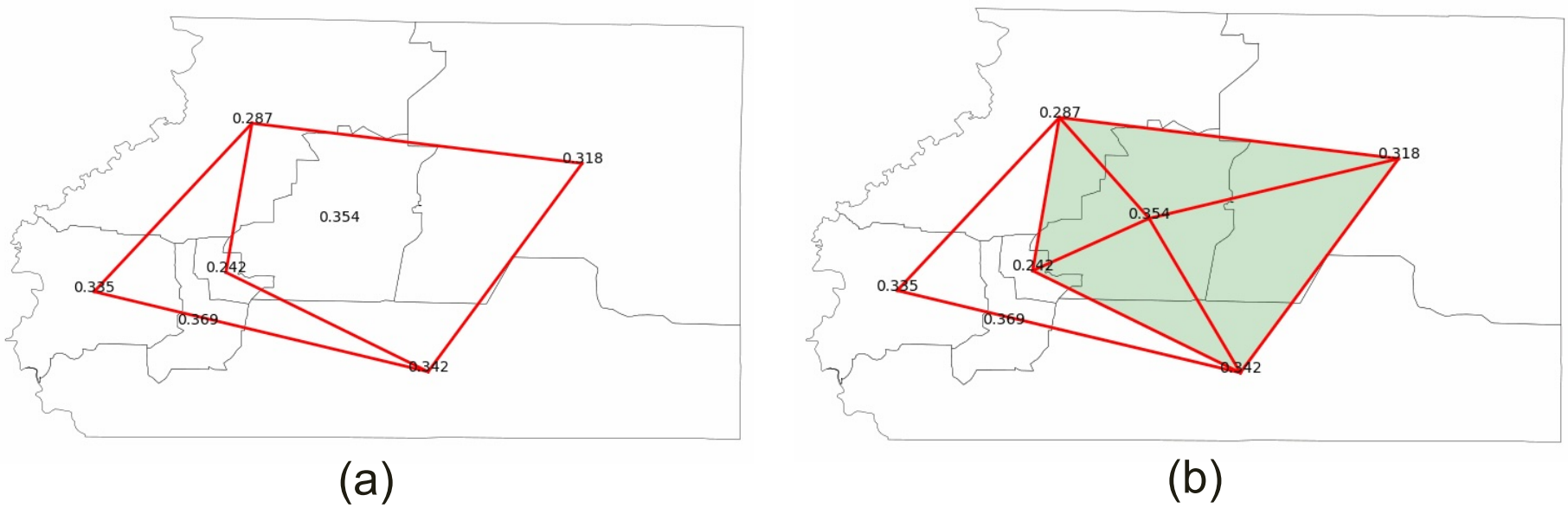
Topological Data Analysis(TDA)



Topological Features



Adjacency Complex

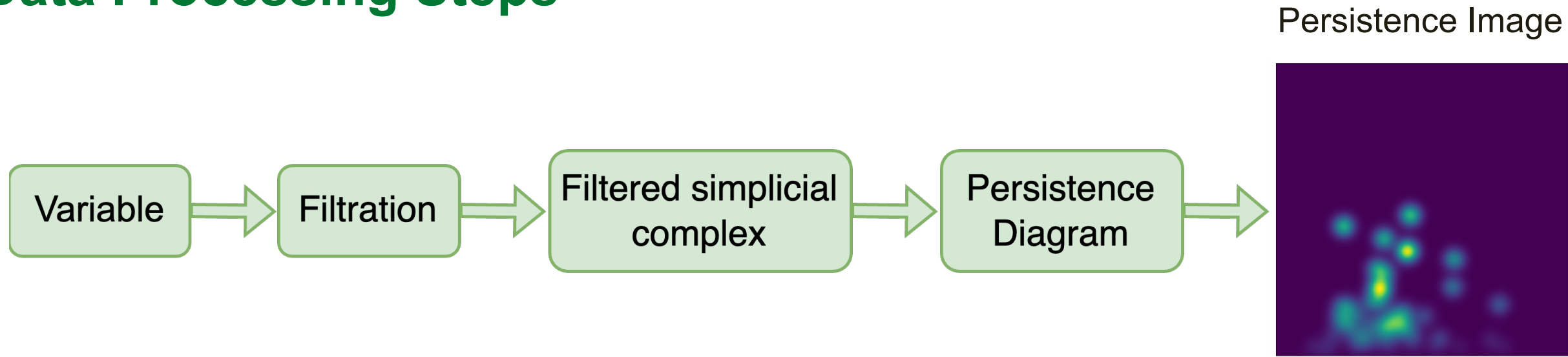


- An adjacency complex is a type of a network constructed based on the adjacency relationship between vertices, i.e., census tracts.
- Adjacency complex captures the topological structure of the data by connecting the vertices that are neighbors.
- Different thresholds for each of the SVI variables create distinct networks.
- We have used each of the SVI components to create different networks.

Acknowledgements

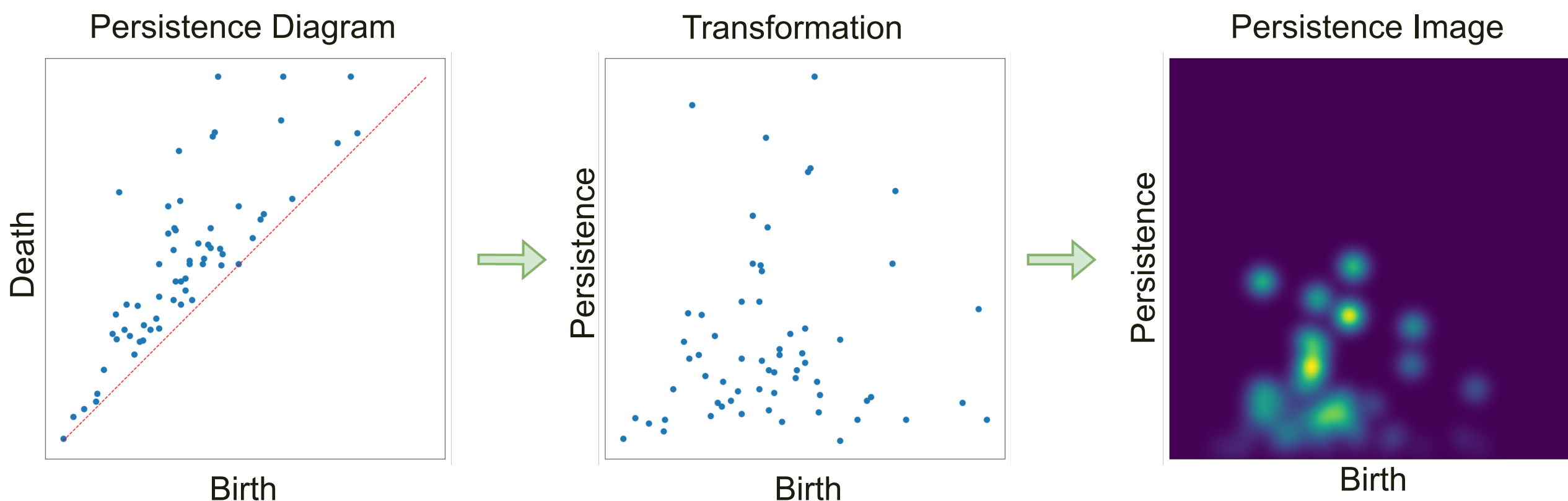
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Data Processing Steps



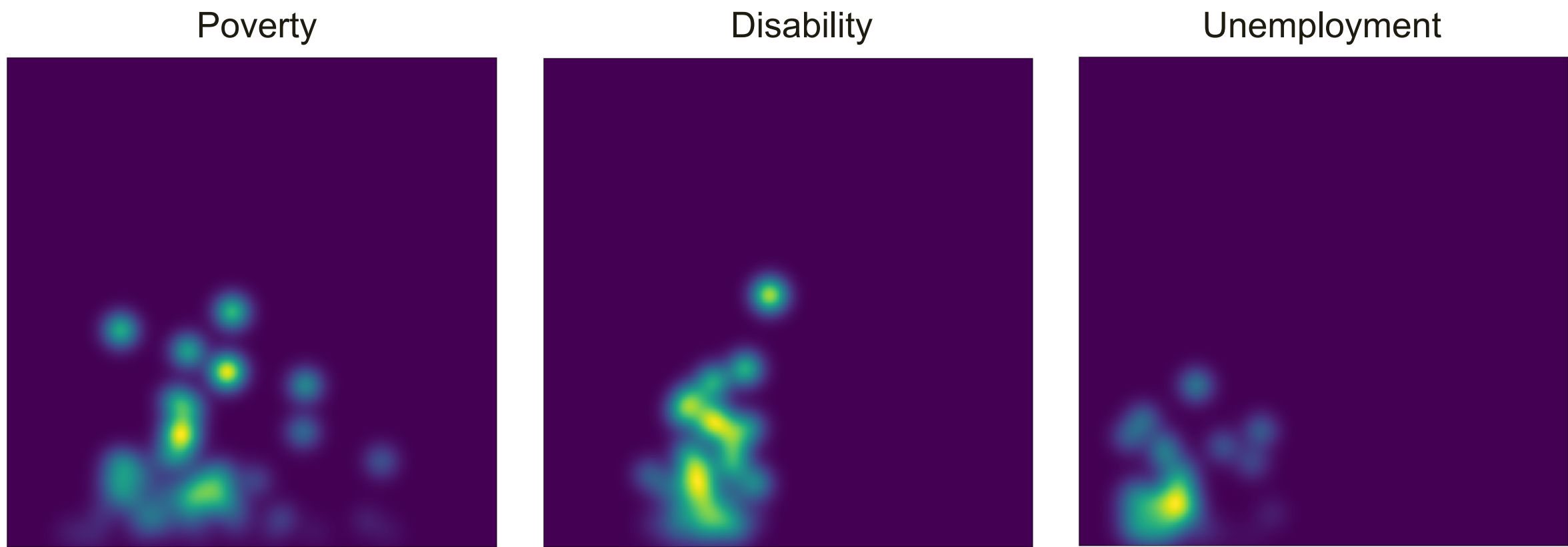
- Each variable in a county generates its own persistence image, which represents topological features.
- For each county, the persistence images are combined into a single matrix to create a unified input for the deep learning model.

Transformation



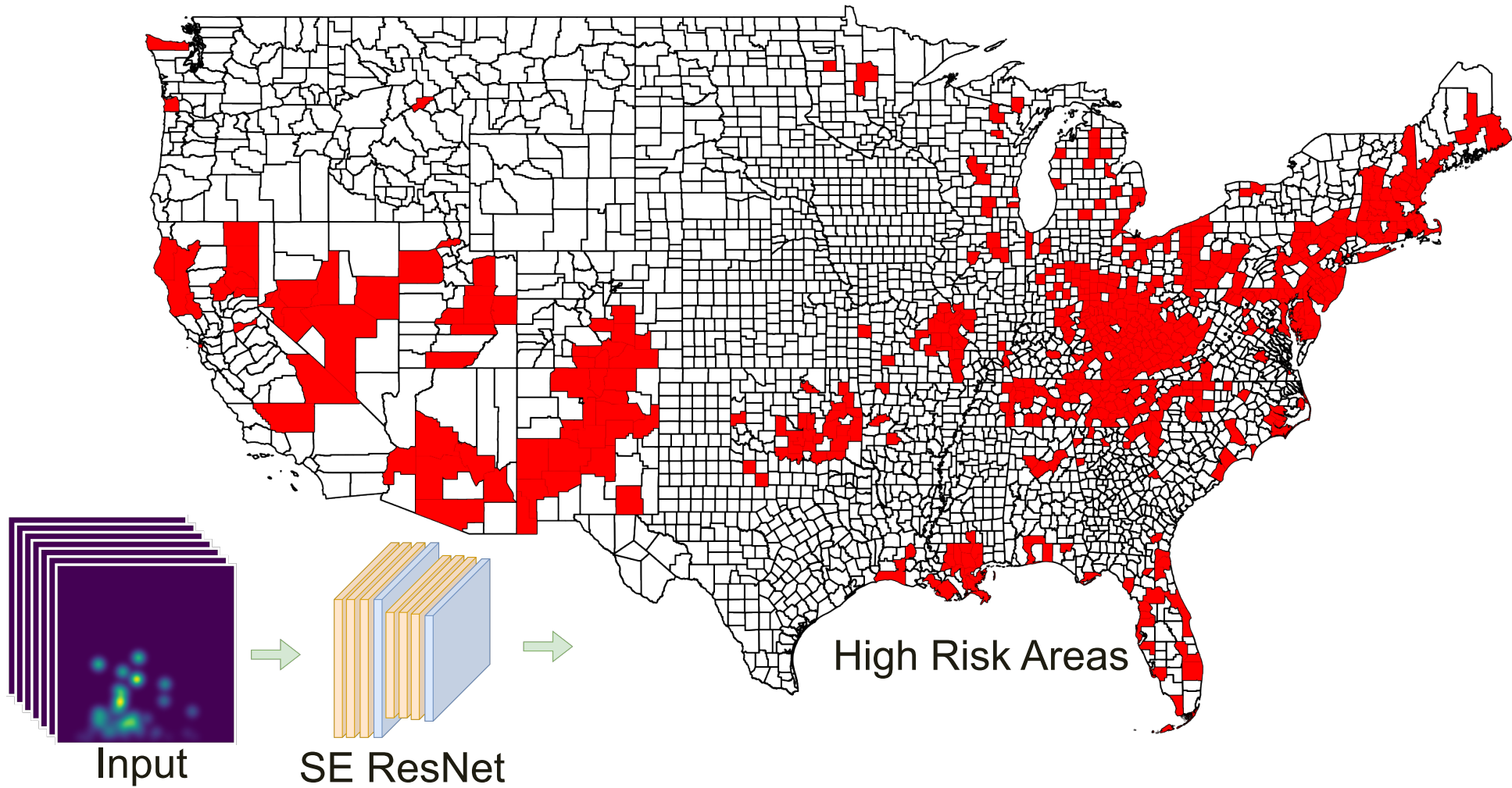
- Persistence diagrams are useful for describing topological features of the data. They are multisets, however and can't be directly used as input for Machine Learning (ML) models.
- To use persistence diagram is in downstream ML tasks we need to suitably vectorize the persistence diagram to a fixed sized representation.
- We map each point in the persistence diagram onto a fixed grid in the birth-persistence plane and then apply a Gaussian filter to create a smooth, continuous image. The resulting persistence image is a fixed size 2D array that captures the topological features of the original data

Persistence Images



- Persistence Images of the Queens county in New York state.

Deep Learning Architecture



- Each county's data is represented as a multi-channel input, consisting of a combination of 16 single-channel persistence images.
- SE-ResNet is employed to classify high-risk areas by adaptively analyzing each channel.

Results

Model	Top 5%	Top 10%	Top 20%
Test Accuracy	90%	75%	65.74%
Test Loss	0.4324	0.5754	0.6186
Low Overdose	TPR: 0.82 FPR:0.05 FNR:0.18	TPR: 0.93 FPR:0.43 FNR:0.07	TPR: 0.92 FPR:0.60 FNR:0.08
High Overdose	TPR: 0.95 FPR:0.18 FNR:0.05	TPR: 0.57 FPR:0.07 FNR:0.43	TPR:0.40 FPR:0.08 FNR:0.60

- Three different models were trained for various risk strata, targeting the top 5%, 10%, and 20% of areas as high overdose risk counties.

Conclusions and Future Directions

- Adding a temporal component to predict the future high-risk overdose areas.
- Improve number features extracted from persistence diagram.
- Use graph neural networks to predict overdose risk.

References:

- Feng, Michelle, and Mason A. Porter. "Persistent homology of geospatial data: A case study with voting." *SIAM Review* 63.1 (2021): 67-99.

For more examples and code, check out our code repository:

